Miguel Saavedra

Mechatronic Engineer

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Personal Details

Birth Date	29th July 1996
Citizenship	Colombia

Education

2020 - 2021 **Postgraduate Diploma in Artificial Intelligence**, *Universidad Autónoma de Occidente*, Cali, Colombia.

2014 - 2019 **BEng (Mechatronics)**, *Universidad Autónoma de Occidente*, Cali, Colombia.
Graduated with High Honour. Weighted GPA of 4.7/5.0

2011 - 2013 **Electronics Technician**, *Institución Educativa Técnico Industrial Rafael Navia Varón*, Cali, Colombia.

Experience

Work Experience

01/2020 - 09/2020 **R&D Robotics Engineer (UAVs)**, *AirflyD & Romero Cano Ingeniería S.A.S.*, Cali, Colombia.
Research and development of a flight controller unit for a heavy-cargo hexacopter with internal combustion engines. The objective is to develop the necessary set of plugins for the accurate control of the aircraft in precision agriculture applications.

02/2019 - 12/2019 **Lead Developer**, *CRT Ingeniería S.A.S. & Romero Cano Ingeniería S.A.S.*, Cali, Colombia.
Developed, tested and implemented software solutions for security systems using deep neural networks and computer vision techniques.
Achievements:
- Developed a low-cost license plate recognition system based on CNN.
- Implemented an image-based heat map algorithm to estimate crowd flow.
- Implemented an image segmentation algorithm for floor detection.

Academic Experience

07/2017 - Present **Member of the Hotbed of Robotics & Autonomous Systems (RAS)**, *Universidad Autónoma de Occidente*, Cali, Colombia.
Developed, tested and implemented different projects as member of RAS. Most of the projects were research initiatives of the university and were presented in local conferences.
Achievements:

Autonomous landing system for an unmanned aerial vehicle on a terrestrial vehicle.

Developed an autonomous landing system for an unmanned aerial vehicle (UAV) on a platform located on the top of a ground vehicle. The system used vision-based techniques to detect the landing platform. Subsequently, the data obtained was processed and used as the process variable for a control system in charge to properly land the UAV on the platform.

Simulation of a landing system for a UAV in Gazebo.

Simulated an autonomous landing system for a UAV with Gazebo, ROS and the Software in the loop provided by PX4. The project consisted in the development of different packages in C++ and Python which allowed the autonomous landing of the UAV.

Detection and tracking of a landing platform for aerial robotics applications.

Developed an embedded vision-based detection and tracking system of a landing platform with ROS and OpenCV. The system extended the capabilities of a popular SURF-based feature detector-descriptor that makes detections of the landing pad alongside a Kalman filter-based estimation module to overcome the issues of only-detector methods.

Mapping and localization in indoors with Turtlebot 2.

Implemented a localization and mapping system (laser-based SLAM) for a turtlebot2 in indoors. The system was capable to accurately localize a robot in a previously mapped environment and subsequently navigate to a specific position in an occupancy grid map.

Projects

11/2019 - 06/2020 **Reinforcement Learning Specialization Project**, *Specialization by University of Alberta on Coursera*.

Trained a lunar lander in a simulated environment (Open AI Gym) with function approximation and control. The agent was implemented with the Expected-Sarsa algorithm and used a Neural network for action-values estimation. The algorithm was capable to do planning steps with experience replay and learn an optimal policy for the landing of the agent.

02/2019 - 05/2019 **Self-Driving Cars Specialization Projects**, *Specialization by University of Toronto on Coursera*.

This work involved four capstone projects in the area of self-driving cars. Topics such as wheeled-robot mathematical modelling through dynamical modelling (tire model), lateral and longitudinal control, state estimation with Kalman filters, visual perception and motion planning were addressed. Most of the projects were developed in Python programming language and tested in the Carla Simulator to assess performance.

Languages

English	Fluent	*IELTS Academic 7.5 overall*
Spanish	Mother-tongue	
French	Basic	

Awards

2014-2019 Academic Excellence Award in Mechatronics Engineering. This scholarship covered the 100% tuition cost during the degree. Granted for an outstanding academic performance and the highest GPA in mechatronics engineering program. Universidad Autónoma de Occidente, Cali, Colombia.

Relevant Courses & certificates

- Reinforcement Learning by University of Alberta & Alberta Machine Intelligence Institute on Coursera. Certificate earned at June 21, 2020
- Self-Driving Cars, a 4-course specialization by University of Toronto on Coursera. Specialization Certificate earned on June 5, 2019

Programming skills

Languages	Python, C, C++, Matlab
Libraries	OpenCV, PyTorch, Scikit learn
Others	ROS, Git, LaTeX, Docker

Publications and poster presentations

M. Saavedra. A. Pinto. V. Romero-Cano. Detection and tracking of a landing platform for aerial robotics applications. IEEE 2nd Colombian Conference on Robotics and Automation CCRA 2018.

M. Saavedra. A. Pinto. V. Romero-Cano. "Development of an autonomous landing system for an unmanned aerial vehicle on a terrestrial vehicle". XXI Encuentro Nacional y XV internacional de Semilleros de Investigación REDCOLSI. October 2018. (Poster Presentation)

M. Saavedra. A. Pinto. V. Romero-Cano. "Localization of a landing platform for an UAV". 5 Encuentro de semilleros de investigación y grupos de estudio de ingeniería mecatrónica y afines. ISSN 2619-4198. 2018. (Poster Presentation)

Research Interests

- Robotics perception, multi-modal sensor fusion, tracking and state estimation
- Artificial intelligence for object detection, classification and segmentation
- Simultaneous localization and mapping (SLAM) and motion planning
- Machine vision, Reinforcement learning

References

Victor Romero-Cano, *PhD in Robotics.*
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David F. Ramirez Moreno, *PhD in Bio-medical Science.*
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Online Information

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